Group Actions on Algebraic Curves
GAAC

Galois Module Structure of groups acting on curves

Research Team:

Karagiannis	**Kostas**
Karakikes	**Miltiadis**
Karanikolopoulos	**Sotiris**
Kontogeorgis	**Aristides**
Lygdas	**Orestis**
Manousoy Sotiropoulou	**Kostantia**
Noulas	**Dimitrios**
Stergiopoulou	**Dimitra-Dionysia**
Terezakis	**Alexis**

The research project is implemented in the framework of H.F.R.I Call “Basic research Financing (Horizontal support of all Sciences)” under the National Recovery and Resilience Plan “Greece 2.0” funded by the European Union Next Generation EU(H.F.R.I. Project Number: 14907.

Contents

1. Group actions of Curves 2
2. Representation Theory 5
3. Modular representations 8
4. Groups with cyclic Sylow subgroup 13
5. De Rham end etale cohomology 14
Summary 16
6. Equivariant Koszul Cohomology of Canonical Curves 17
References 18

1. Group actions of Curves

1.1. **Algebraic Curves.** By an algebraic curve we mean a projective, non-singular, one-dimensional variety defined over an algebraically closed field k of characteristic $p \geq 0$. Over the field $\mathbb{C}$ of complex numbers, the notion of an algebraic curve coincides with that of a compact Riemann surface. Every compact Riemann surface X is an orientable two-dimensional real manifold, and to any such surface we can associate a natural number $g \in \mathbb{N}$, called the genus, which topologically counts the number of holes of the surface X. Over an arbitrary field of positive characteristic we can still define the genus as

$$g = \dim_k H^0(X, \Omega_X),$$

although a topological interpretation is less clear.

An automorphism of a curve X is an isomorphism $\sigma : X \to X$, and all automorphisms form a group under composition. If the genus is zero, then X is isomorphic to $\mathbb{P}^1$, and the automorphism group is the group of Möbius transformations $\mathrm{PGL}(2, k)$, which is infinite. If $g = 1$, then the curve X admits a group structure and acts on itself by translations, so $X \subset \mathrm{Aut}(X)$, and hence $\mathrm{Aut}(X)$ is infinite as well.

When $g \geq 2$, the automorphism group is finite. This was a classical result in the case of Riemann surfaces, where a proof follows by studying the action of the automorphism group on the Weierstrass points of the curve. For curves in positive characteristic, a suitable notion of Weierstrass points and the finiteness of the automorphism group were first established by F. K. Schmidt [64] using the theory of Hasse derivatives.

Moreover, it is known that for any finite group G there exists a curve X such that $\mathrm{Aut}(X) \cong G$ (see [47]). Note that most curves have trivial automorphism group [51], [59], since curves with non-trivial automorphisms correspond to singularities of the moduli space of curves of genus g. Finding explicit examples of curves without automorphisms is not easy (see [58]).

The automorphism group of a curve of genus $g \geq 2$ is finite. However, for fixed genus, the size of the groups that can appear is bounded. In characteristic zero, using the Riemann-Hurwitz formula and a case-by-case analysis, Hurwitz proved that for $g \geq 2$

$$|\mathrm{Aut}(X)| \leq 84(g-1). \tag{1.1}$$

1.2. **Ramification Filtration.** In order to explain the situation in characteristic $p > 0$, we need to introduce the ramification filtration at a closed point $x \in X$. Let G be a subgroup of $\mathrm{Aut}(X)$ and let $m_{X,x}$ be the maximal ideal of the local ring $\mathcal{O}_{X,x}$. We denote by $k(x)$ the residue field of x. For $i \geq 0$, the i^{th} lower ramification subgroup $G_{x,i}$ of G at x is the subgroup of all elements $\sigma \in G$ which fix x and act trivially on $\mathcal{O}_{X,x}/m_{X,x}^{i+1}$. These groups form a decreasing finite sequence

$$G_{x,0} \rhd G_{x,1} \rhd \cdots \rhd G_{x,n} \rhd G_{x,n+1} = \{1\}. \tag{1.2}$$

When the characteristic $p = 0$, it is known that $G_{x,1} = \{1\}$. In general, $G_{x,0}/G_{x,1}$ is a cyclic group of order prime to p, while for $i \geq 1$, $G_{x,i}/G_{x,i+1}$ is an elementary abelian group, i.e. a group isomorphic to the direct sum of finitely many cyclic groups of order p. If $G_{x,1} = \{1\}$ for every $x \in X$, then the cover $X \to X/G$ is said to be *tame*, otherwise it is called *wild*. If $G_{x,2} = \{1\}$, then the ramification is called *weak*.

The Riemann-Hurwitz formula [69, Sec. 3.4, p. 90] relates the genera of the curves $X, Y = X/G$ as follows:

$$2(g_X - 1) = 2(g_Y - 1)|G| + \sum_{x \in X} \sum_{i=1}^{\infty} \big(|G_{x,i}| - 1\big). \tag{1.3}$$

For tame covers, the Hurwitz bound remains the same. For general wild ramified curves, H. Stichtenoth [67, 68] proved that the following bound holds:

$$|\mathrm{Aut}(X)| \leq 16 g_X^4, \tag{1.4}$$

with the Hermitian Fermat curve as the only exception, given by

$$0 = x^{p^h+1} + y^{p^h+1} + z^{p^h+1} = \underline{x} \cdot \left(\underline{x}^t\right)^{p^h}, \qquad \underline{x} = (x, y, z). \tag{1.5}$$

Notice that for $h = 0$ the above Fermat curve is a quadratic form, while for $h > 0$ it behaves as a Frobenius-shifted quadratic form and has $\mathrm{PGU}(3, p^{2h})$ as automorphism group [45, 38].

The result of H. Stichtenoth was improved by H. Henn [25], who proved that

$$|\mathrm{Aut}(X)| \leq 8 g_X^3, \tag{1.6}$$

with a finite list of exceptions. Henn's result contained a gap which was filled by M. Giulietti and G. Korchmáros (see [19]).

All exceptions in Henn's list involve curves whose automorphism group has a large p-subgroup compared to its genus. C. Lehr and M. Matignon [44] introduced the notion of a *big action*, namely when $\mathrm{Aut}(X)$ contains a p-subgroup P such that

$$|P| > \frac{2p}{p-1} g_X. \tag{1.7}$$

M. Matignon and M. Rocher [50, 62, 61] and M. Giulietti and G. Korchmáros [18] studied and classified big actions defined by equations similar to (1.7).

In characteristic $p > 0$, the p-rank of the Jacobian γ_X plays a role similar to the rank of the homology group; in fact $0 \leq \gamma_X \leq g_X$. Curves with $g_X = \gamma_X$ are called *ordinary* and they form a Zariski-dense set in the moduli space of curves of genus g. For such curves, S. Nakajima [57] proved the bound

$$|\mathrm{Aut}(X)| \leq 84(g_X - 1) g_X. \tag{1.8}$$

He further observed that his bound could not be best possible and, by studying the Artin-Schreier-Mumford curve

$$(x^{p^h} - x)(y^{p^h} - y) = c, \qquad c \in k, \tag{1.9}$$

he conjectured that the optimal bound is given by a cubic polynomial in $\sqrt{g_X}$.

Understanding the automorphism group is an interesting problem on its own and has many applications to counting points, moduli problems etc.

1.3. **Representation filtration.** In [32] A. Kontogeorgis and S. Karanikolopoulos introduced the notion of representation filtration for the study of jumps for Harbater–Katz–Gabber covers or HKG-covers for short.

More precisely HKG-covers, which are defined as Galois covers of the projective line with a unique wildly ramified point and at most one tamely ramified point. These covers are significant because they serve as minimal compactifications of local actions on complete local rings, playing a crucial role in deformation theory and lifting problems.

- Compactification: HKG-covers serve as minimal compactifications of Galois actions on complete local rings (actions on $k[[t]]$),. This allows us to attach global invariants, such as genus, differentials, and p-rank, to local actions.
- Zero p-Rank: The curve X in an HKG-cover has zero p-rank. In fact, a curve X with a p-group $G \subseteq \mathrm{Aut}(X)$ forms an HKG-cover $X \to X/G$ if and only if X has zero p-rank and the order of the group $|G|$ is a pole number at the unique stabilized point.
- Deformation Theory: These covers are utilized in the study of lifting problems and the deformation theory of curves with automorphisms.

A connection between the classical ramification filtration and the pole numbers at P is given by the following observation

- *Ramification Filtration:* Defined by the stabilizer subgroups $G_i(P)$ acting trivially on $O_P/\mathfrak{m}_P^{i+1}$.
- Weierstrass Semigroup $H(P)$: The numerical semigroup of pole numbers $n \in \mathbb{N}$ for functions $f \in F$ with $(f)_\infty = nP$.

To bridge these concepts, *representation filtration* is introduced based on the kernels of representations $\rho_i : G_1(P) \to \mathrm{GL}(L(m_i P))$, where m_i are pole numbers.

Key Theorems on Filtrations For HKG-covers:

(1) The jumps of the ramification filtration coincide with the jumps of the representation filtration. Specifically, if n is the number of representation jumps, there are exactly n ramification jumps.
(2) The ramification groups G_{b_i} are equal to the representation kernels $\ker \rho_{c_i}$ for all jumps.
(3) The ramification jumps are explicitly characterized in terms of pole numbers m_ν such that $m_r - m_\nu$ is a jump, where m_r is the smallest pole number not divisible by p.

Structure of the Weierstrass Semigroup The generators of the Weierstrass semigroup $H(P)$ using the jumps of the representation filtration can be described.

- For every representation jump c_i, there exists a generator of $H(P)$ of the form $\bar{m}_i = p^{h_i}\lambda_i$ where $(\lambda_i, p) = 1$.
- The Weierstrass semigroup $H(P)$ for these covers is always symmetric, meaning the maximum gap is $2g_X - 1$. The structure depends on the second ramification group $G_2(P)$:
- If $G_1(P) > G_2(P)$, the extension is HKG, $|G_2(P)|$ equals the first pole number m_1, and the minimal generators are derived entirely from the representation jumps.
- If $G_1(P) = G_2(P)$, the order $|G_1(P)|$ is required as an additional generator for $H(P)$.

Applications and Zero p-Rank

- **Zero p-rank:** A curve X has zero p-rank and $|G|$ is a pole number at the stabilized point if and only if the cover $X \to X/G$ is an HKG-cover.
- **Big Actions:** The results apply to "big actions" (where $|G| > \frac{2g_X}{p-1}p$), showing that for such curves, $|G_1(P)|$ is not a minimal generator of $H(P)$.
- **Hasse-Arf Theorem:** For abelian $G_1(P)$, the Hasse-Arf divisibility conditions can be reinterpreted in terms of congruences involving the generators of the Weierstrass semigroup.

Holomorphic Polydifferentials for HKG-covers Bases for the spaces of polydifferentials $H^0(X, \Omega_X^{\otimes m})$ are related to functions associated with pole numbers. This basis is used to prove the symmetry of the Weierstrass semigroup and to analyze the indecomposability of the module under the action of $G_1(P)$.

1.4. **Mumford Curves.** One can observe that X can be uniformized by a discrete subgroup Γ of $\mathrm{PSL}(2,\mathbb{R})$, i.e. $X \cong \Gamma\backslash\mathbb{H}$, and the Hurwitz upper bound given in eq. (1.1) is equivalent to Siegel's lower bound $\pi/21$ on the volume of the fundamental domain of a Fuchsian group ([14, exer. 6 p. 245],[43]).

Let K be a non-archimedean valued field. D. Mumford [52] showed that curves defined over K, whose stable reduction is split multiplicative, i.e. a union of rational curves intersecting at $\bar{K}$-rational points, are isomorphic to an analytic space of the form $\Gamma\backslash(\mathbb{P}^1_K - \mathcal{L}_\Gamma)$, where Γ is a discontinuous group in $\mathrm{PGL}(2,K)$ and $\mathcal{L}_\Gamma$ is the set of limit points. The automorphism group of the curve X is then isomorphic to the group N/Γ, where N equals the normalizer of $\Gamma \in \mathrm{PGL}(2,K)$, [12],[17, p. 216].

We will call such curves *Mumford curves*. Notice that not all curves defined over K admit such a uniformisation. For example the Artin-Schreier Mumford curve has split multiplicative reduction and is a Mumford curve only if $|c| < 1$. The uniformization theory can give stronger results when applied to Mumford curves.

Herrlich [26] has shown that for p-adic Mumford curves of genus $g \geq 2$ and $p \geq 7$ the Hurwitz bound can be strengthened to $12(g_X - 1)$.

Notice that by the work of Manin-Drinfeld [48] and Gerritzen [16], Mumford curves are known to be ordinary, therefore the Nakajima bound given in eq. (1.8) holds. For Mumford curves defined over non-archimedean fields of positive characteristic G. Cornelissen, F. Kato and the PI [12] proved that Nakajima's conjecture was correct for Mumford curves and the following bound holds:

$$|\mathrm{Aut}(X)| \leq \max\left\{12(g_X - 1), 2\sqrt{g}(\sqrt{g}+1)^2\right\}. \tag{1.10}$$

They also classified those curves for which $|\mathrm{Aut}(X)| \geq 12(g_X - 1)$. Moreover the above bound is best possible since it is attained for the Artin-Schreier-Mumford curves given by eq. (1.9).

This theorem can also be reformulated in the style of Siegel lower bound as follows: the $\mu(N)$ invariant [33, eq. 2] of its normalizer N is bounded from below by

$$\mu(N) \geq \min\left\{1/12, \frac{\sqrt{g}-1}{2\sqrt{g}(\sqrt{g}+1)}\right\}.$$

Notice that $\mu(N)$ plays the role of of a Gauss-Bonnet "volume" and the index $[N : \Gamma]$ which equals the order of automorphism group can be evaluated in terms of theorems of HNN groups as in theorem 2 in [33].

Concerning the Nakajima conjecture for ordinary curves X over an algebraically closed field of characteristic $p > 0$, R. Guralnik and M. Zieve in a Workshop in Leiden on Automorphisms of curves in 2004 announced that there exists a sharp bound of the order of $g^{8/5}$ for $|\mathrm{Aut}(X)|$.

For groups with a specific structure we can have better bounds. For example S. Nakajima in [56] used the Hasse-Arf theorem in order to prove that

$$|\mathrm{Aut}(X)| \leq 4g_X + 4,$$

and this bound can be further improved for abelian automorphisms groups by V. Rotger and the second author in [39], to the bound

$$|\mathrm{Aut}(X)| \leq 4(g_X - 1).$$

2. Representation Theory

Consider the space $M_1 := H^0(X, \Omega_X)$ of holomorphic differentials of X, acted on by the automorphism group G of X. In characteristic zero the knowledge of the Galois module structure of M_1 is a classical result due to Hurwitz [29]. In positive characteristic if the cover $X \to X^G$ is unramified or $(|G|, p) = 1$ then the Galois module structure of M_1 is known due to results of Tamagawa [70]. Valentini [72] generalised the result for unramified extensions with p-groups as Galois groups. Moreover Salvador and Bautista [46] determined the semisimple part of the representation with respect to the Cartier operator in the case of p-group. For the cyclic group case Valentini and Madan [73] and members of our research team [31] determined the structure of M_1 in terms of indecomposable modules. The same study has been done for the elementary abelian case by Calderón, Salvador and Madan [63]. For an application of K-theory to this problem, see [5]. Finally, the P.I. in a joint article with F. Bleher and T. Chinburg proved that if the group G has non trivial cyclic p-Sylow subgroups, then the Galois module structure depends on the ramification type and on the fundamental characters attached on closed ramified points, [2].

2.1. **Explicit Methods.** A strategy for studying the $k[G]$-module structure is to first write explicit bases of the spaces $H^0(X, \Omega_X^{\otimes m})$.

2.1.1. *Boseck Theory.* Usually, a curve with a non-trivial automorphism group comes with a natural Galois cover $\pi : X \to X/H = Y$, where Y is a known curve (usually $\mathbb{P}^1$ or an elliptic curve) and H is a subgroup of the full automorphism group G. In this way the divisor of the H-invariant differential $\mathrm{div}\pi^* dx$ can be computed in terms of the pullback formula [23, prop. 2.3 p. 301]

$$\mathrm{div}\pi^* dx \cong \pi^* \mathrm{div}(dx) + R_{X/Y}, \tag{2.1}$$

where $R_{X/Y}$ denotes the ramification divisor of the cover.

Once the divisor $\mathrm{div}(dx)$ is computed, finding the space of holomorphic (poly)differentials is the same as computing the Riemann-Roch space $L(\mathrm{div} dx)$. This method was used by H. Boseck in [6], who gave precise formulas for both Kummer and Artin-Schreier extensions of the projective line. Once a basis is constructed one has to identify the indecomposable summands. For the case of cyclic group action the last computation essentially is equivalent to the computation of the Jordan normal form.

This method was used in [73], [31] and also by articles of Rzedowski-Calderón, Villa-Salvador and Madan [63] and Marques and Ward [49] for some other groups under additional hypotheses on the cover $X \to X/G$.

2.1.2. *Mumford Curves.* For the case of Mumford curves there is a pure group theoretic approach to the determination of global sections of holomorphic differentials initiated by the work of V. Drinfeld and Y. Manin [48]. For holomorphic polydifferentials there is also a group theoretic approach, the theory of harmonic measures studied by J. Teitelbaum and P. Schneider see [65], [71].

Let K be a complete non-archimedean valued field, $\Gamma \subseteq \mathrm{PGL}(2, K)$ is a Schottky subgroup, and X_Γ the Mumford curve obtained from Γ. N denotes the normalizer of Γ in $\mathrm{PGL}(2, K)$, the quotient group $G = N/\Gamma$, which acts on X_Γ from the left, is the automorphism group $\mathrm{Aut}(X_\Gamma)$ of X_Γ over K. Recall that Γ is a free group of finite rank, whose rank, say g, is equal to the genus of X_Γ. Let us fix a free generating set $\{\gamma_1, \ldots, \gamma_g\}$ of Γ. For any right $K[\Gamma]$-module P, each derivation $d\colon \Gamma \to P$ is uniquely determined by its values $h_i = d(\gamma_i)$ for $1 \leq i \leq g$, and conversely, since Γ is free, such values $h_i \in P$ can be freely chosen to obtain a derivation d; indeed, once h_i's are chosen, then $d(w)$ for any $w \in \Gamma$ is uniquely determined by the derivation rules.

For a positive integer n, we consider the $2n - 1$ dimensional vector space of polynomials $P_n \subseteq K[T]$ of degree $\leq 2(n-1)$. The group $\mathrm{PGL}(2, K)$ acts on P_n from the right as follows: for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{PGL}(2, K)$ and $F \in P_n$, we define

$$F^\gamma(T) := \frac{(cT + d)^{2(n-1)}}{(ad - bc)^{n-1}} F\left(\frac{aT + b}{cT + d}\right) \in K[T]. \tag{2.2}$$

Now, consider the $(2n-1)g$-dimensional space $\mathrm{Der}(\Gamma, P_n)$ of derivations, which can be seen as an N-module as follows: for $\delta \in N$ and $d \in \mathrm{Der}(\Gamma, P_n)$, define

$$(d^{\delta})(\gamma) = [d(\delta\gamma\delta^{-1})]^{\delta} \tag{2.3}$$

for $\gamma \in \Gamma$. There is then a well defined $G = N/\Gamma$ action on the group cohomology $H^1(\Gamma, P_n)$, since Γ acts trivially modulo principal derivations.

Theorem 1 ([71, Theorem 1]). *For any $n \geq 1$, the space $H^0(X_\Gamma, \Omega_{X_\Gamma}^{\otimes n})$ of n-differentials on the curve X_Γ is naturally isomorphic to the space group cohomology $H^1(\Gamma, P_n)$. Moreover, this identification is G-equivariant with respect to the natural right G-action on $H^0(X_\Gamma, \Omega_{X_\Gamma}^{\otimes n})$.* □

In [40] this approach is used in order to study the $K[G]$-module structure of polydifferentials for the case of Artin-Screier-Mumford curves, where $N = A * B$, $\Gamma = [A, B]$ and $A, B \subset \mathrm{PGL}(2, K)$ are cyclic groups of order p generated by

$$\epsilon_A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \epsilon_B = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}, \tag{2.4}$$

respectively, where $s \in K^\times$ and $|s| > 1$.

2.2. **Fermat Curves.** In [8], using the known explicit bases for holomorphic differentials for the Fermat curves the structure of the canonical ring $R = \bigoplus_{m=0}^{\infty} H^0(F_n, \Omega_{F_n}^{\otimes m})$ of a Fermat curve $F_n : x^n + y^n + z^n = 0$ under the action of its automorphism group $G = (\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}) \times S_3$ is determined. The primary contribution is the explicit calculation of the classes of the graded components $[H^0(F_n, \Omega_{F_n}^{\otimes m})]$ in the Grothendieck group $K_0(G, K)$ for all degrees $m \geq 1$. This decomposition relies on computing the characters of the irreducible representations of G and the characters of the module of holomorphic m-differentials, followed by an inner product computation. The authors establish a canonical basis for the m-differentials using a quotient construction $W_m/I_m \cong H^0(F_n, \Omega_{F_n}^{\otimes m})$ that is well-suited for group actions, where W_m and I_m are vector spaces generated by lattice points in a triangle $E_{m(n-3)}$. The multiplicities obtained are then used to determine the equivariant Hilbert series $H_{R,G}(t)$ as an explicit rational function, providing a condensed description of the infinite-dimensional G-action. The final formulas are implemented in a Sage program, allowing for the computational analysis of any arbitrary Fermat curve, and the results are verified by showing that the sum of the equivariant Hilbert series yields the expected non-equivariant Hilbert series.

2.3. **The Chevalley-Weyl Theory.** The study of the Galois module structure of the space $H^0(X, \Omega_X)$ is a classical problem first posed by Hecke [9] and solved by, Chevalley and Weil [10] using character theory. In [13] Ellingsrud and Lønsted generalized the results of Chavalley-Weil by introducing an equivariant Lefschetz formula for finite reductive groups acting on projective varieties, focusing on the case of smooth projective curves, with a finite reductive group acting on them which essentially means that the group G has order prime to the characteristic of the field.

Finite Groups Acting on Schemes and Sheaves. Let us now introduce the necessary technical framework and notation for describing group actions in algebraic geometry, which are crucial for the main problem of determining the Galois structure of holomorphic differentials, see [53], [22] .

Let X be a Noetherian scheme, locally separated over a field k, and let H be a finite group acting faithfully on X on the right over k. The action is denoted by $\vartheta : X \times_k H \longrightarrow X$.

A core concept is that of a quasi-coherent $\mathcal{O}_X$-H-module $\mathcal{F}$, also known as a quasi-coherent H-sheaf or H-equivariant sheaf. This is a quasi-coherent sheaf of $\mathcal{O}_X$-modules together with an isomorphism $\phi : \vartheta^*\mathcal{F} \longrightarrow p_1^*\mathcal{F}$ of $\mathcal{O}_{X\times_k H}$-modules, which must satisfy an associativity (cocycle) condition on $X \times_k H \times_k H$. Equivalently, this $\mathcal{O}_X$-H-module has a compatible action of H on its stalks, requiring $h(\alpha \cdot f) = h(\alpha) \cdot h(f)$ for appropriate local sections α and f. If $\mathcal{F}$ is coherent or locally free coherent as an $\mathcal{O}_X$-module, it is called a coherent or locally free coherent $\mathcal{O}_X$-H-module, respectively. An important example is the sheaf of relative differentials Ω_X, which is a coherent $\mathcal{O}_X$-H-module.

The existence of the quotient scheme $Z = X/H$ requires that the H-orbit of every point in X be contained within an affine open subset, a condition always met if X is quasi-projective. If the quotient $\pi : X \to Y = X/I$ exists for a subgroup $I \leq H$, the pushforward $\pi_*\mathcal{F}$ is a quasi-coherent $\mathcal{O}_Y$-module. When I is a normal subgroup of H and $\mathcal{J}$ is an H-conjugation-stable ideal of the group ring $k[I]$, the sheaf $\mathcal{K}$ defined as the kernel of $\mathcal{J}$ acting on $\pi_*\mathcal{F}$ is shown to be a quasi-coherent $\mathcal{O}_Y$-H-module.

Let us now describe the Ellingsrud-Lønsted approach. The G-Lefschetz trace of a G-linearized coherent sheaf $\mathcal{F}$ on a projective variety X with a linear algebraic group action G is defined in the representation ring $R_k(G)$ of G over the base field k as:

$$L_{G,X}(\mathcal{F}) = \sum (-1)^i \mathrm{Tr}(G|H^i(X,\mathcal{F})).$$

Their article provides various expressions for $L_{G,X}(\mathcal{F})$, depending on the nature of $\mathcal{F}$, which involve numerical invariants of $\mathcal{F}$, X, $Y = X/G$, and a local contribution from the ramification locus of the morphism $X \to Y$. These formulas are viewed as equivariant versions of the Hirzebruch-Riemann-Roch formula, or as Lefschetz formulas. One version generalizes the Chevalley-Weil formula for the representation of G on the m-differentials in the complex case (Theorem 3.8).

The proof uses equivariant K-theory and the Hirzebruch-Riemann-Roch formula for locally free sheaves on a curve, replacing the transcendental constructions and matriciel Riemann-Roch formula used in the original Chevalley-Weil proof. The computations are inspired by similar work in the topological case by Atiyah-Bott-Segal-Singer related to the index theorem. The base field k is assumed to be algebraically closed with arbitrary characteristic.

Equivariant K-Theory and the Lefschetz Trace. $R_k(G)$ is the representation ring of G over k. $K_G(X)$ is the Grothendieck group of coherent $G-\mathcal{O}_X$-Modules, and $K_G^{\cdot}(X)$ is for locally free $G-\mathcal{O}_X$-Modules.

The category $\underline{\mathrm{Mod}}^f_{G,X}$ consists of $G-\mathcal{O}_X$-Modules $\mathcal{F}$ with finite-dimensional cohomology groups. The Grothendieck group of this subcategory is $K_G(X)^f$.

Definition of the Lefschetz Trace For an object $\mathcal{F}$ in $\underline{\mathrm{Mod}}^f_{G,X}$, the Lefschetz trace is defined as:

$$L_{G,X}(\mathcal{F}) = \sum_{i=0}^{\infty} (-1)^i [H^i(X,\mathcal{F})].$$

$L_{G,X}$ extends to a group homomorphism $L_{G,X} : K_G(X)^f \to R_k(G)$. If X is proper over k, then $K_G(X)^f = K_G(X)$.

The Lefschetz Formula in the Unramified Case. Let G be a finite, reductive group acting on an algebraic k-scheme X, such that the quotient scheme $Y = X/G$ exists, and $\pi : X \to Y$ is the canonical projection. Recall that a finite reductive group G is one where the order of G, $|G|$, is not divisible by $\mathrm{char}(k)$.

Ellingsrud Lønsted gave a general formula for $L_{G,X}(\mathcal{F})$ in terms of the Euler characteristic of certain associated sheaves on the quotient scheme Y. Let $\hat{G}$ be a complete set of representatives of the irreducible G-k-modules. Setting $\mathcal{F}_W = \pi_*^G(\mathcal{F} \otimes_k W)$, where $\pi_*^G(\mathcal{F}) = (\pi_*(\mathcal{F}))^G$, the following is shown:

$$L_{G,X}(\mathcal{F}) = \sum_{V \in \hat{G}} \chi(\mathcal{F}_{\tilde{V}}) \cdot [V].$$

If the morphism $\pi : X \to Y$ is étale (e.g., G acts freely on X), then for all $G-\mathcal{O}_X$-Modules $\mathcal{F}$, the Lefschetz formula is:

$$L_{G,X}(\mathcal{F}) = \frac{1}{|G|} \cdot \chi(\mathcal{F}) \cdot [k[G].$$

Assuming X is a smooth connected projective curve over k with a G-action, the paper specializes the formula for a locally free $G-\mathcal{O}_X$-Module $\mathcal{E}$. We denote by g the genus of X, g' the genus Y, $n = |G|$, $Q_1, \ldots, Q_t \in Y$ are the ramification points, and e_Q is the ramification index at Q.

Using the Riemann-Roch formula $\chi(\mathcal{E}) = \deg(\mathcal{E}) + \mathrm{rk}\,\mathcal{E} \cdot (1-g)$, the length of the ramification residue module $l(\mathcal{R}(\mathcal{E} \otimes V))$, and Lemma 2.2, the first precise formula is derived:

$$L_{G,X}(\mathcal{E}) = \left(\frac{1}{n} \cdot \deg \mathcal{E} + (1-g')\mathrm{rk}\,\mathcal{E}\right)[k[G]] - \frac{1}{n} \sum_{V \in \hat{G}} l(\mathcal{R}(\mathcal{E} \otimes \tilde{V})) \cdot [V].$$

By locally analyzing the contribution of the ramification points $Q \in Y$ and the decomposition group $G_P \simeq \mathbb{Z}/e_Q$ at a branch point $P \in \pi^{-1}(Q)$, the local length term is expressed using the multiplicity $n_{d,Q,\mathcal{E},V}$ of the character $\chi^d_{e_Q}$ in $V_{\mathcal{E},P} \otimes_k V$. This leads to the reformulation:

$$L_{G,X}(\mathcal{E}) = \left(\frac{1}{n} \cdot \deg \mathcal{E} + (1-g')\mathrm{rk}\,\mathcal{E}\right)[k[G]] - \sum_{V \in \hat{G}} \sum_{Q} \sum_{d=1}^{e_Q} \left(1 - \frac{d}{e_Q}\right) n_{d,Q,\mathcal{E},V}[V].$$

Chevalley-Weil Generalization (Theorem 3.8) For an invertible $G - \mathcal{O}_X$-Module $\mathcal{L}$, $L_{G,X}(\mathcal{L})$ is generalized to the Chevalley-Weil formula:

$$L_{G,X}(\mathcal{L}) = \left(\frac{1}{n} \cdot \deg \mathcal{L} + (1-g') - \sum_Q \left(1 - \frac{1}{e_Q}\right)\right)[k[G]] + \sum_V \sum_Q \sum_{d=0}^{e_Q-1} \left\langle \frac{r_Q - 1 - d}{e_Q} \right\rangle n_{d,Q,V} \cdot [V],$$

where $\chi_{e_Q}^{r_Q}$ is the character by which G_P acts on $\mathcal{L}_P$, $n_{d,Q,V} = n_{d,Q,\mathcal{O}_X,V}$ is the multiplicity of $\chi_{e_Q}^d$ in V, and $\langle x \rangle = x - [x]$ is the fractional part.

Further Formulas.

Proposition 4.2. For a locally free $G - \mathcal{O}_X$-Module $\mathcal{E}$ and an invertible $\mathcal{O}_Y$-Module $\mathcal{M}$ (with X a smooth, connected, projective curve), the following is established:

$$L_{G,X}(\mathcal{E} \otimes \pi^*\mathcal{M}) = L_{G,X}(\mathcal{E}) + \deg \mathcal{M} \cdot \operatorname{rk} \mathcal{E} \cdot [k[G]]$$

Corollary 4.4: Equivariant Hilbert-Samuel Polynomial. Setting $\mathcal{E}(m) = \mathcal{E} \otimes \pi^*\mathcal{M}^{\otimes m}$, the Lefschetz trace $L_{G,X}(\mathcal{E}(m))$ is a polynomial of degree one in m, with coefficients in $R_k(G)$. The leading coefficient, $\deg \mathcal{M} \cdot \operatorname{rk} \mathcal{E} \cdot [k[G]]$, is called the degree of $\mathcal{E}$ relative to $\pi^*\mathcal{M}$.

In [36] B. Köck provides a new approach to calculating the *equivariant Euler characteristic* $\chi(G, X, \mathcal{E})$ of *Zariski sheaves* on smooth projective curves, generalizing the classical Chevalley-Weil formula.

In [36, th. 1.1] a formula for the equivariant Euler characteristic $\chi(G, X, \mathcal{E}) = [H^0(X, \mathcal{E})] - [H^1(X, \mathcal{E})]$ is given, where $\mathcal{E}$ is a locally free G-sheaf and $\pi : X \to Y = X/G$ is a tamely ramified cover.

$$n \cdot \chi(G, X, \mathcal{E}) = (n(1 - g_Y)r + \deg(\mathcal{E}))[k[G]] - \sum_{P \in X} \sum_{d=0}^{e_P-1} d \left[\operatorname{Ind}_{G_P}^G(\mathcal{E}(P) \otimes \chi_P^d)\right],$$

where r is the rank of $\mathcal{E}$ and $n = |G|$.

The approach of B. Köck provides very natural and quick proof of the classical Chevalley-Weil formula proved by Ellingsrud and Lønsted, which is based on on the Lefschetz fixed point formula. Also this approach is extended to the computation of equivariant Euler characteristic for étale sheaves, proving an equivariant Grothendieck-Ogg-Shafarevich formula. This formula includes terms for the wild conductor $\alpha(\mathcal{F})$, covering cases beyond the strictly tamely ramified scenario implicitly covered by the original Ellingsrud-Lønsted formula.

In summary, Köck confirms the Ellingsrud-Lønsted formula via a new, more algebraic method based on the Lefschetz formula and proceeds to solve the analogous problem in étale cohomology.

Applications. In [7] we study the same problem as in [8] but now using the Ellingsrud- Lønsted approach. We were investigating the problem of decomposing finite group actions (G) on graded modules, specifically the canonical ring $S_X = \bigoplus_{m=0}^{\infty} H^0(X, \Omega_X^{\otimes m})$ of a projective curve X. The main results include general formulas [8, th. 3] and an algorithm [8, Algorithm 2] for computing the *equivariant Hilbert series* $H_{S_X,V}(T)$ for an irreducible representation V of G. This approach utilizes the Chevalley-Weil formula to express the multiplicities $N_{V,m}$ of V in each graded piece $W_m = H^0(X, \Omega_X^{\otimes m})$ in terms of the ramification data of the cover $\pi : X \to X/G$. Unlike traditional methods which require computing explicit bases for polydifferentials, here we compute ramification indices (e_Q), the genus of the quotient curve (g_Y), and multiplicities $n_{d,Q,V}$ over the decomposition groups G_P. In this way the equivariant Hilbert series for *Fermat curves* F_n (with affine model $x^n + y^n + 1 = 0$) as a specific application of their general formula (Theorem 4) is determined. Furthermore, Petri's Theorem is employed, which provides a short exact sequence $0 \to I_X \to S \to S_X \to 0$ relating the canonical ring S_X to the symmetric algebra S. This way we compute the equivariant Hilbert series for the *canonical ideal* I_X by subtraction: $H_{I_X,V}(T) = H_{S,V}(T) - H_{S_X,V}(T)$. All these computations, including the implementation of Molien's formula for $H_{S,V}(T)$, are provided in a Sage program.

3. Modular representations

When studying the action of automorphism groups on the cohomology of algebraic or topological objects, the passage from ordinary to modular representation theory introduces substantial complications. Over characteristic 0, the cohomology groups decompose into direct sums of irreducible representations, and the associated character theory provides a complete and transparent description of the action: eigenvalues, traces, and multiplicities are directly accessible through standard tools. In the modular setting, however, cohomology groups often carry representations that are no longer semisimple, and extension classes between composition factors become essential to understanding the action. Projective modules and their covers intervene naturally, and the relationship between ordinary and modular characters—reflected in decomposition numbers—obscures the direct extraction of invariants such as fixed-point dimensions or Lefschetz numbers. Moreover, the block structure of the automorphism group interacts nontrivially with the grading or weight structure of the cohomology, so that even the determination of composition factors requires subtle local information. As a consequence, phenomena that are straightforward in characteristic 0, such as detecting symmetries or determining the contribution of various automorphisms to cohomological invariants, may become significantly more intricate in the modular framework.

Nakajima's Results on Galois Module Structure of Cohomology Groups. The work of S. Nakajima [54], [55] investigates the module structure of cohomology groups $H^i(X, f^*(\mathcal{F}))$ when $f : X \to Y$ is a finite Galois covering of projective varieties with Galois group $G = \mathrm{Gal}(X/Y)$, and $\mathcal{F}$ is a coherent sheaf on Y. This analysis is conducted without assumption on the characteristic of the base field k.

General Results for Arbitrary Dimension (Unramified Case). For a finite étale Galois covering, the primary general result is given by [54, Thm 1]:

(1) There exists a finite complex of $k[G]$-modules $(L) : 0 \to L^0 \to L^1 \to \cdots \to L^m \to 0$ such that each L^i is a finitely generated free $k[G]$-module.
(2) The i-th cohomology group of the complex (L) is isomorphic to $H^i(X, f^*(\mathcal{F}))$ as a $k[G]$-module.

This complex provides a "relation" among the $k[G]$-modules $H^i(X, f^*(\mathcal{F}))$. A key consequence of this complex structure is [54, Th. 2]: if $H^i(X, f^*(\mathcal{F})) = 0$ for all indices i except for a single value $i = n$, then the remaining cohomology group $H^n(X, f^*(\mathcal{F}))$ is a free $k[G]$-module.

Results for Curves (Unramified Case). When X and Y are connected complete non-singular curves and $\mathcal{F} = \mathcal{K}$ is the canonical sheaf of Y, the $k[G]$-module structure of the space of holomorphic differentials $\mathfrak{D} = H^0(X, f^*(\mathcal{K}))$ is fully determined by the exact sequence given in Theorem 4:

$$0 \to \mathfrak{D} \to k[G]^g \to I_G \to 0,$$

where g is the genus of Y, and I_G is the augmentation ideal of $k[G]$. This sequence uniquely determines $\mathfrak{D}$ up to isomorphism.

- As a corollary, if $k[G]$ is semi-simple (e.g., $\mathrm{char}(k) \nmid |G|$), the sequence splits, and $\mathfrak{D} \cong k \oplus k[G]^{g-1}$, which generalizes the classical result of Chevalley and Weil.
- In the case where $p = \mathrm{char}(k) > 0$ and G is a p-group, $\mathfrak{D}$ is decomposed into indecomposable modules as $\mathfrak{D} \cong \Omega^2 k \oplus k[G]^{g-d}$, where $\Omega^2 k$ is the second loop space of the trivial module k, and d is the minimal number of generators of G.

3.1. Extension to Tamely Ramified Coverings

Nakajima later extended these results to the case where $f : X \to Y$ is a tamely ramified finite Galois covering.

(1) *General Result (Arbitrary Dimension):* For a tamely ramified covering, the equivalent of [54, Thm. 1] holds, see [55, Thm. 1]. A complex (L) of finitely generated projective $k[G]$-modules exists whose cohomology is isomorphic to $H^i(X, \mathcal{F})$.
(2) *Key Corollary:* If $H^i(X, \mathcal{F}) = 0$ for all i except $i = n$, then $H^n(X, \mathcal{F})$ is a projective $k[G]$-module.
(3) *Curves (Dimension 1):* For non-singular curves and a locally free G-sheaf $\mathcal{E}$, [55, Thm. 2] provides a "relation" between $H^0(X, \mathcal{E})$ and $H^1(X, \mathcal{E})$. This relation involves auxiliary projective $k[G]$-modules L^0 and L^1 in an exact sequence:

$$0 \to H^0(X, \mathcal{E}) \to L^0 \to L^1 \to H^1(X, \mathcal{E}) \to 0$$

and specifies a stable isomorphism between $L^0 \oplus r \cdot N$ and $L^1 \oplus E$. The module N is determined by the covering $f : X \to Y$ and reflects the ramification state, and E is determined by the local $k[G_P]$-module structure of $\mathcal{E}$ at the ramified points P.

A Generalization of Euler-Poincaré characteristic. In [11] a detailed discussion and generalization of the Euler-Poincaré characteristics is given and a the classical coherent Euler-Poincaré characteristic is lifted to a more refined invariant $\chi R\Gamma^+(f_*(T))$ that lies in a special Grothendieck group associated with Galois module structure.

Let $f : X \to Y$ be a tame G-covering of schemes proper and of finite type over a noetherian ring A. Let T be a coherent G-sheaf on X (or a bounded complex of such sheaves).

The traditional, or naive, Euler-Poincaré characteristic is the alternating sum of the classes of cohomology groups in the common Grothendieck group of all finitely generated AG-modules, $G_0(AG)$

$$\chi(G,T) = \sum_i (-1)^i \cdot [H^i(X,T)].$$

The refined equivariant Euler-Poincaré characteristic $\chi R\Gamma^+(f_*(T))$ as a class in $CT(AG)$, the Grothendieck group of all finitely generated cohomologically trivial AG-modules.

$$\chi R\Gamma^+(f_*(T^\bullet)) = \sum_{i=-\infty}^{\infty} (-1)^i \cdot [M^i],$$

where $M^\bullet$ is a bounded complex of finitely generated, cohomologically trivial AG-modules that is isomorphic to $R\Gamma^+(f_*(T^\bullet))$ in the derived category $D^+(A,G)$. This invariant is a canonical lift of $\chi(G,T)$.

The refined Euler-Poincaré characteristic, $\chi R\Gamma^+(f_*(T))$, is a lift of the naive Euler-Poincaré characteristic, $\chi(G,T)$, in the sense that it represents the same "content" but within a smaller, more tightly constrained group where elements capture more structural information. This concept of lifting is formalized through the relationship between two specific algebraic structures called Grothendieck groups:

The Naive Characteristic $\chi(G,T)$**:** This is a class in the Grothendieck group $G_0(AG)$. The group $G_0(AG)$ consists of classes of all finitely generated AG-modules. On the other hand $\chi(G,T)$ is defined as the alternating sum of the classes of the cohomology groups: $\chi(G,T) = \sum_i (-1)^i \cdot [H^i(X,T)]$.

The Refined Characteristic $\chi R\Gamma^+(f_*(T))$**:** This is a class in the Grothendieck group $CT(AG)$. The group $CT(AG)$ is defined as the Grothendieck group of all finitely generated cohomologically trivial AG-modules. A module M is cohomologically trivial if its Tate cohomology groups vanish for all subgroups $H \subset G$. The relationship is established by the natural forgetful homomorphism:

$$\nu : CT(AG) \longrightarrow G_0(AG).$$

The map ν takes the class of a cohomologically trivial module in $CT(AG)$ and "forgets" that it has the special cohomologically trivial property, mapping it to its class in the more general $G_0(AG)$. The term lift means that the element $\chi R\Gamma^+(f_*(T))$ in $CT(AG)$ maps down precisely to $\chi(G,T)$ in $G_0(AG)$ via the forgetful map ν.

The existence of a canonical lift is significant because $CT(AG)$ is, in general, neither surjective nor injective with respect to $G_0(AG)$. By coming from $CT(AG)$, the naive characteristic $\chi(G,T)$ is confined to a particular subset of $G_0(AG)$, thereby restricting the possible values $\chi(G,T)$ can take.

Since $CT(AG)$ is a more constrained group, elements within it inherently carry more detailed, subtle, and profound structural information about the underlying AG-modules than elements in $G_0(AG)$ do. For example, the invariant $\Psi(X/Y)$ defined using this structure recovers the stable isomorphism class of the ring of integers, which is a subtle arithmetic invariant.

The Role of Numerical Tameness. The existence and computability of $\chi R\Gamma^+(f_*(T))$ rely crucially on a generalized notion of tameness called *numerical tameness*.

A G-covering $f : X \to Y$ is numerically tame, if for every point $y \in Y$, a local flat base change $Y' \to Y$ exists such that the base-changed cover $X \times_Y Y' \to Y'$ is G-equivariantly isomorphic to a quotient:

$$X \times_Y Y' \simeq (Z \times_{Y'} G_{Y'})/H,$$

where $Z \to Y'$ is an H-covering for a group H whose order is prime to the residue characteristic of Y'. This "numerical" condition ensures that the local Galois module structure is simple.

For a numerically tame G-covering $f : X \to Y$ and a quasicoherent G-sheaf T on X, the following conditions for the direct image sheaf $F = f_*(T)$ are equivalent and satisfied:

- All stalks of F are cohomologically trivial AG-modules.
- For every open affine subset U of Y, $F(U)$ is a cohomologically trivial AG-module.

This property guarantees that the terms of the Cech hypercohomology complex used to compute $R\Gamma^+(f_*(T))$ consist of cohomologically trivial modules, which is essential for the construction of the final complex $M^\bullet$

Motivation and Generalizations. The work in this area synthesizes ideas from two classical areas of Galois module theory:

- Galois Structure of Differentials (Geometric): Initiated by Hecke and generalized by Chevalley and Weil, this deals with the determination of the G-module structure (multiplicity of characters) in the space of holomorphic differentials $H^0(X, \Omega^1)$ on a compact Riemann surface.
- Galois Structure of Rings of Integers (Arithmetic): This focuses on the determination of the stable isomorphism class of the ring of integers $\mathcal{O}_N$ in a tame Galois extension N/K, a problem famously solved by Taylor's theorem linking the class to root numbers of L-series.

The refined invariant $\chi R\Gamma^+(f_*(T))$ is designed to connect these fields, leading to conjectures and theorems that generalize Taylor's result to tame G-coverings of schemes, especially smooth projective varieties over a finite field.

Weak ramification. B. Köck in [35] gives a description of the *Galois module structure* of the Zariski cohomology groups $H^i(X, \mathcal{E})$, for $i = 0, 1$, where $\mathcal{E}$ is a G-equivariant locally free sheaf on X is given. Köck's paper shows that weak ramification is the precise condition under which the Galois module structure of Zariski cohomology behaves almost as well as in the tame case. The results combine local ramification theory, modular representation theory, and equivariant Riemann-Roch techniques, and they provide a conceptual framework for understanding wild ramification in the cohomology of curves.

Let X be a smooth projective curve over an algebraically closed field k of characteristic $p > 0$, and let $G \subset \mathrm{Aut}(X)$ be a finite group. Denote by $\pi : X \longrightarrow Y := X/G$ the quotient map.

The novelty of this work lies in extending classical results (Chevalley-Weil, Kani, Nakajima) from the tamely ramified case to the more general *weakly ramified* situation, where wild ramification is allowed but restricted.

For a point $P \in X$, let $G_{P,s}$ denote the s-th ramification group. The cover π is said to be *weakly ramified* if

$$G_{P,s} = \{1\} \quad \text{for all } s \geq 2 \text{ and all } P \in X.$$

Equivalently, wild ramification may occur, but only in the simplest possible form.

Weak ramification is a natural condition: for example, if X is an ordinary curve and G is arbitrary, then π is weakly ramified by the Deuring-Shafarevich formula.

Normal Basis Theorem for Fractional Ideals. A key local ingredient is a refinement of Noether's normal basis theorem.

Theorem 2 (Normal bases for fractional ideals)**.** *Let L/K be a finite Galois extension of local fields with Galois group G. For an integer $b \in \mathbb{Z}$, the fractional ideal $\mathfrak{m}_L^b$ is free of rank one over $\mathcal{O}_K[G]$ if and only if:*

- *L/K is weakly ramified, and*
- *$b \equiv 1 \pmod{|G_1|}$, where G_1 is the first ramification group.*

This result is crucial: it provides a precise criterion for when local pieces of a G-equivariant line bundle behave well as G-modules.

Projectivity of Zariski Cohomology. Let $D = \sum_P n_P[P]$ be a G-invariant divisor on X. Köck proves a sharp characterization of when the cohomology of $\mathcal{O}_X(D)$ is projective as a $k[G]$-module.

Theorem 3. *Suppose $\pi : X \to Y$ is weakly ramified and*

$$n_P \equiv -1 \pmod{e_P^w} \quad \text{for all } P \in X,$$

where $e_P^w = |G_{P,1}|$ is the wild part of the ramification index. Then:

(1) The equivariant Euler characteristic

$$\chi(G, X, \mathcal{O}_X(D)) = [H^0(X, \mathcal{O}_X(D))] - [H^1(X, \mathcal{O}_X(D))]$$

lies in the image of the Cartan homomorphism $K_0(k[G]) \to K_0(G, k)$.

(2) If one of the groups H^0 or H^1 vanishes, the other is a projective $k[G]$-module.

Conversely, if $\deg D > 2g_X - 2$ and $H^0(X, \mathcal{O}_X(D))$ is projective, then π must be weakly ramified and the above congruence conditions hold.

This result generalizes Nakajima's theorem from tame to weak ramification.

Equivariant Euler Characteristic Formula. A central result of [35] is an explicit formula for the equivariant Euler characteristic of a locally free G-sheaf.

Let $\mathcal{E}$ be a locally free G-module of rank r on X. For each $P \in X$, let χ_P denote the character of G_P acting on the cotangent space $\mathfrak{m}_P/\mathfrak{m}_P^2$.

Theorem 4 (Equivariant Riemann-Roch). *In $K_0(G,k) \otimes \mathbb{Q}$ one has*

$$\chi(G, X, \mathcal{E}) = c\,[k[G]] - \frac{1}{|G|} \sum_{P \in X} \sum_{d=1}^{e_P^t - 1} e_P^w\, d \left[\mathrm{Ind}_{G_P}^G \left(\mathcal{E}(P) \otimes \chi_P^d\right)\right],$$

where e_P^t is the tame part of the ramification index and c is an explicit constant depending on $\deg \mathcal{E}$, g_Y, and the ramification data.

This formula extends the Chevalley-Weil and Ellingsrud-Lønsted formulas to arbitrary (possibly wildly ramified) covers, though only at the level of Grothendieck groups.

The Ramification Module. In the weakly ramified case, Köck constructs a canonical projective $k[G]$-module $N_{G,X}$ encoding the global contribution of wild ramification.

Theorem 5. *There exists a unique projective $k[G]$-module $N_{G,X}$ such that*

$$|G| \cdot N_{G,X} \;\cong\; \bigoplus_{P \in X} \bigoplus_{d=1}^{e_P^t - 1} \bigoplus^{e_P^w d} \mathrm{Ind}_{G_P}^G \left(\mathrm{Cov}(\chi_P^d)\right),$$

where $\mathrm{Cov}(\chi_P^d)$ denotes the projective cover of χ_P^d. Moreover,

$$[N_{G,X}] = (1 - g_Y)[k[G]] - \chi(G, X, \mathcal{O}_X(E)),$$

with $E = \sum_P (e_P^w - 1)[P]$.

This module generalizes the ramification modules of Kani and Nakajima from the tame case.

Applications. Two notable applications are given:
- A description of $H^1(X, \mathcal{I}(S))$, where $\mathcal{I}(S)$ is the ideal sheaf of a G-stable set S containing all wildly ramified points.
- A generalization of Kani's theorem: if S contains all ramified points, then
$$H^0(X, \Omega_X(S)) \oplus N_{G,X}$$
is a free $k[G]$-module.

In [37] the results of [35] are used in order to compute the dimension of the tangent space of the equivariant deformation functor associated with a finite p-group G acting on a smooth projective curve X in characteristic p.

3.2. **Equivariant Deformation Theory.** There is the following connection between the dimension of the tangent space of the deformation functor and equivariant quadratic differentials:

$$\dim_k H^1(G, \mathcal{T}_X) = \dim_k H^0(X, \Omega_X^{\otimes 2})_G,$$

where
- $\dim_k H^1(G, \mathcal{T}_X)$: The dimension of the tangent space of the equivariant deformation functor D. This dimension is a measure of the degrees of freedom in deforming X while preserving the G-action.
- $\dim_k H^0(X, \Omega_X^{\otimes 2})_G$: The dimension of the space of coinvariants of G acting on $V = H^0(X, \Omega_X^{\otimes 2})$, the space of global holomorphic quadratic differentials on X.

The article provides explicit formulas for $\dim_k H^0(X, \Omega_X^{\otimes 2})_G$ based on the structure of the group G and the nature of its action. Let $Y = X/G$ be the quotient curve, g_Y be its genus, and P_j for $j = 1, \ldots, r$ be the representatives for the G-orbits of ramified points.

(1) Cyclic Case (G is a Cyclic p-Group): This result is derived from a theorem by Borne on the Galois module structure of Riemann-Roch spaces, which uses indecomposable $k[G]$-modules V_l.

$$\dim_k H^0(X, \Omega_X^{\otimes 2})_G = 3g_Y - 3 + \sum_{j=1}^{r} \left\lfloor \frac{2d(P_j)}{e_0(P_j)} \right\rfloor,$$

where $d(P_j)$ is the coefficient of the ramification divisor at P_j, and $e_0(P_j)$ is the ramification index.

(2) Weakly Ramified Case (Second Ramification Group Vanishes) The action is weakly ramified if the second ramification group $G_2(P)$ is trivial for all points $P \in X$.

$$\dim_k H^0(X, \Omega_X^{\otimes 2})_G = 3g_Y - 3 + \sum_{j=1}^{r} \log_p |G(P_j)| + \begin{cases} 2r & \text{if } p > 3 \\ r & \text{if } p = 2 \text{ or } 3 \end{cases}$$

This formula is derived by showing that an auxiliary space of meromorphic quadratic differentials W is a free $k[G]$-module.

p-Rank Representation. The space $H^0(X, \Omega_X^{\otimes 2})$ is a $k[G]$-module that decomposes into a semisimple part H_D^s (the p-rank representation) and a nilpotent part H_D^n. H_D^s is shown to be a free $k[G]$-module, and its rank is computed based on the p-rank of Y (γ_Y) and the ramification points.

The formula for the rank of H_D^s depends on the genus of the quotient Y:

$$b(G, D, k) = \begin{cases} \gamma_Y - 1 + r & \text{if } g_Y = 0 \\ \gamma_Y - 1 + r + \deg(\mathrm{div}(\phi)_{\mathrm{red}}) & \text{if } g_Y \geq 1 \end{cases}$$

where ϕ is a specific holomorphic differential on Y.

4. Groups with cyclic Sylow subgroup

In [2] the Galois structure of the space of holomorphic differentials of curves is studied. Let X be a smooth projective geometrically irreducible curve over a perfect field k of positive characteristic p. The space of holomorphic differentials of X is $H^0(X, \Omega_X)$, which is a left $k[G]$-module if G is a finite group acting faithfully on X over k. We would like to know the decomposition of $H^0(X, \Omega_X)$ into its indecomposable direct $k[G]$-module summands, especially if p divides the order of the group G.

If G is a group with a non-cyclic p-Sylow subgroup, the set of indecomposable $k[G]$-modules is infinite. If, moreover, $p > 2$ then the indecomposable $k[G]$-modules are considered impossible to classify (cf. [60]). On the other hand side finite groups with a cyclic p-Sylow subgroup have a finite set of indecomposable $k[G]$-modules (cf. [27], [4], [24]).

If the group G has a cyclic p-Sylow subgroup, then the $k[G]$-module structure of $H^0(X, \Omega_X)$ is uniquely determined by the lower ramification groups $G_{x,i}$ and the fundamental characters θ_x of closed points x of X that are ramified in the cover $X \longrightarrow X/G$.

Using the Conlon induction theorem, the problem is reduced to determining the structure over p-hypo-elementary subgroups $H \subset G$, which are semidirect products $H = P \rtimes C$, where P is a normal cyclic p-subgroup and C is a cyclic p'-group.

For a p-hypo-elementary group H, let I be a normal cyclic p-subgroup of H. A filtration is defined on the sheaf $\pi_*\Omega_X$ using powers of the Jacobson radical of $k[I]$. The key technical step is showing that the quotient sheaves $\mathcal{L}_j = \Omega_X^{(j+1)}/\Omega_X^{(j)}$ are line bundles on $Y = X/I$, specifically:

$$\mathcal{L}_j \cong S_{\chi^{-j}} \otimes_k \Omega_Y(D_j),$$

where $S_{\chi^{-j}}$ is a one-dimensional $k[H]$-module, and D_j is an effective H-invariant divisor determined explicitly by the ramification data of the cover $X \longrightarrow X/I$.

The isomorphism of $k[H/I]$-modules:

$$H^0(X, \Omega_X)^{(j+1)}/H^0(X, \Omega_X)^{(j)} \cong H^0(Y, \mathcal{L}_j)$$

is established. Since $Y \longrightarrow X/H$ is tamely ramified, the $k[H/I]$-module structure of $H^0(Y, \Omega_Y(D_j))$ is determined by the ramification data of $X \longrightarrow X/H$ using earlier results by Nakajima and Kani. This information, combined with the classification of indecomposable modules for p-hypo-elementary groups, determines the $k[H]$-module structure of $H^0(X, \Omega_X)$.

A similar result was proved by F. Bleher and A. Wood in [3] for holomorphic polydifferentials.

$k[G]$-module structure of de Rham cohomology. In [15] the Galois module structure of the de Rham cohomology of an algebraic curve in positive characteristic p, is determined for groups with a cyclic p-Sylow subgroup.

More precisely for a group G with a cyclic p-Sylow subgroup acting on X, defined over a perfect field k of characteristic p, the $k[G]$-module structure of $H^1_{dR}(X)$ is uniquely determined by the ramification data of the cover $X \to X/G$ and the genus of X. The ramification data includes:

- The stabilizers of the action on $X(\overline{k})$.

- The lower (or equivalently, upper) ramification jumps of the action corresponding to the stabilizers.
- The fundamental characters of the action.

In the case $G = \mathbb{Z}/p^n$ and $\pi : X \to Y$ is a $\mathbb{Z}/p^n$-cover over an algebraically closed field k, then the following explicit result is established: The de Rham cohomology $H^1_{dR}(X)$ is isomorphic as a $k[\mathbb{Z}/p^n]$-module to:

$$H^1_{dR}(X) \cong J_{p^n}^{\oplus 2(g_Y-1)} \oplus J_{p^n-p^{n-m}+1} \oplus \bigoplus_{Q\in B} J^{\oplus 2}_{p^n-p^n/e_Q} \oplus \bigoplus_{Q\in B} \bigoplus_{t=0}^{m_Q-1} J^{\oplus 2}_{p^n-p^n/u_Q^{(t+1)}},$$

where J_i denotes the indecomposable $k[G]$-module given by the Jordan block of size i (since G is a p-group, J_i is uniquely determined by its size)g_Y is the genus of $Y = X/G$, $m = \max\{m_{X/Y,P}\}$ is the maximal power of p dividing the ramification index,B is the ramification locus, e_Q is the ramification index, and $u_Q^{(t)}$ are the upper ramification jumps at a ramified point Q.

5. De Rham end etale cohomology

In [66] the Galois module structure of the $H^1(X_{\text{et}}, \mathbb{F}_p)$ is introduced and studied as a $k[G]$-module, where G is a subgroup of the automorphism group.

5.1. **Characteristic zero.** In characteristic zero, there is a close relation between $H^1_{\text{dR}}(X)$, the homology group $H_1(X,\mathbb{Z})$ and the space of holomorphic differentials $H^0(X,\Omega_X)$ as $\mathbb{C}[G]$-modules. This technique was used in [34] in order to compute and compare the Galois module structure of homology for a cyclic cover of the projective line with homological techniques and by using the Chevalley-Weil formula.

More precisely, the space of holomorphic differentials $\Omega^1(X) = H^0(X,\Omega^1_X)$ on a compact Riemann surface X of genus g (where $\dim_{\mathbb{C}} \Omega^1(X) = g$) is isomorphic as a $\mathbb{C}$-vector space to the $\mathbb{C}$-vector space $H^1(X,\mathbb{C})$. First Serre duality provides a natural isomorphism:

$$H^1(X,\mathcal{O}_X) \cong \Omega^1(X)^*,$$

where $\Omega^1(X)^*$ is the dual space of $\Omega^1(X)$. For a compact Riemann surface X, the *Hodge Principle* and the *De Rham Isomorphism* yield the relation:

$$H^1_{dR}(X,\mathbb{C}) \cong H^0(X,\Omega^1_X) \oplus H^1(X,\mathcal{O}_X).$$

Since $H^1_{dR}(X,\mathbb{C}) \cong H^1(X,\mathbb{C}) \cong H_1(X,\mathbb{C})^*$ (where $H_1(X,\mathbb{C}) = H_1(X,\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{C}$), we have the *Hodge Decomposition*:

$$H^1(X,\mathbb{C}) \cong \Omega^1(X) \oplus \overline{\Omega^1(X)}, \tag{5.1}$$

($\overline{\Omega^1(X)}$ is the space of anti-holomorphic 1-forms, which is isomorphic to $H^1(X,\mathcal{O}_X)$). Equation (5.1) is also a decomposition of $\mathbb{C}[G]$-modules, that is the character of $H^1(X,\mathbb{C})$ is:

$$\chi_{H^1(X,\mathbb{C})} = \chi_{\Omega^1(X)} + \overline{\chi_{\Omega^1(X)}}.$$

Thus, the $\mathbb{C}[C]$-module structure of $H^1(X,\mathbb{C})$ (which is the dual of $H_1(X,\mathbb{C})$) is completely determined.

Let us write for the $\mathbb{C}[G]$-module structure of the homology is:

$$H_1(X,\mathbb{C}) \cong \bigoplus_{\nu=0}^{n-1} M_\nu \cdot \chi_\nu$$

where M_ν is the multiplicity of χ_ν in $H^1(X,\mathbb{C})$. Due to the Hodge decomposition, the multiplicity M_ν is given by:

$$M_\nu = \text{mult}(\chi_\nu, \Omega^1(X)) + \text{mult}(\chi_\nu, \overline{\Omega^1(X)}) = m_\nu + m_{n-\nu},$$

where m_ν is the multiplicity of χ_ν in $\Omega^1(X)$ (from the Chevalley-Weil type formula).

5.2. **Characteristic p.** Let X be an irreducible, smooth, and complete curve over an algebraically closed field k of characteristic $p > 0$. Let G be a finite subgroup of $\mathrm{Aut}(X)$.

There is no direct isomorphism between

$$H^1_{\mathrm{dR}}(X) \qquad \text{and} \qquad H^1_{\text{ét}}(X, \mathbb{F}_p) \otimes_{\mathbb{F}_p} k,$$

but there is a precise and well-understood relationship mediated by the Cartier operator, the Hodge-de Rham filtration, and the Artin-Schreier sequence.

For a smooth proper curve X/k, $H^1_{\mathrm{dR}}(X)$ has dimension $2g$ and fits into the Hodge filtration

$$0 \longrightarrow H^0(X, \Omega^1_X) \longrightarrow H^1_{\mathrm{dR}}(X) \longrightarrow H^1(X, \mathcal{O}_X) \longrightarrow 0.$$

In characteristic p, this sequence need not split canonically nor equivariantly.

The Frobenius morphism induces a p-linear operator, the Cartier operator

$$C : H^0(X, \Omega^1_X) \longrightarrow H^0(X, \Omega^1_X),$$

whose kernel and image control much of the structure.

Étale cohomology with $\mathbb{F}_p$-coefficients is governed by the Artin-Schreier exact sequence

$$0 \longrightarrow \mathbb{F}_p \longrightarrow \mathcal{O}_X \xrightarrow{F-\mathrm{id}} \mathcal{O}_X \longrightarrow 0.$$

Taking cohomology yields

$$0 \longrightarrow H^1_{\text{ét}}(X, \mathbb{F}_p) \longrightarrow H^1(X, \mathcal{O}_X) \xrightarrow{F-\mathrm{id}} H^1(X, \mathcal{O}_X).$$

Hence:

$$\boxed{H^1_{\text{ét}}(X, \mathbb{F}_p) \;\cong\; \ker\big(F - \mathrm{id} : H^1(X, \mathcal{O}_X) \to H^1(X, \mathcal{O}_X)\big)}$$

Tensoring with k,

$$H^1_{\text{ét}}(X, \mathbb{F}_p) \otimes_{\mathbb{F}_p} k \;\subset\; H^1(X, \mathcal{O}_X).$$

Cartier theory gives a canonical isomorphism

$$H^1_{\text{ét}}(X, \mathbb{F}_p) \otimes k \;\cong\; \ker(C : H^0(X, \Omega^1_X) \to H^0(X, \Omega^1_X)),$$

dual to the above description via Serre duality. Thus:

- Étale H^1 with $\mathbb{F}_p$-coefficients corresponds to Cartier-invariant classes
- De Rham H^1 contains these classes, but is larger

The precise relation is:

$$\boxed{H^1_{\text{ét}}(X, \mathbb{F}_p) \otimes k \;\cong\; \ker(C) \;\subset\; H^0(X, \Omega^1_X) \;\subset\; H^1_{\mathrm{dR}}(X)}$$

Key consequences:

- There is no natural isomorphism between $H^1_{\mathrm{dR}}(X)$ and $H^1_{\text{ét}}(X, \mathbb{F}_p) \otimes k$
- Étale $\mathbb{F}_p$-cohomology captures only the Cartier-stable part
- Equality holds only in very special cases (e.g. ordinary curves, after choosing splittings)
- $H^1_{\mathrm{dR}}(X)$: differential-geometric / crystalline information
- $H^1_{\text{ét}}(X, \mathbb{F}_p)$: arithmetic, inseparable phenomena
- Cartier operator bridges the two worlds

Crystalline approach.

Setup. Let k be a perfect field of characteristic $p > 0$, and let X/k be a smooth, proper, geometrically connected curve of genus g. We write $W(k)$ for the ring of Witt vectors of k.

All cohomology theories below are taken with respect to X.

Crystalline cohomology. The first crystalline cohomology group

$$H^1_{\mathrm{cris}}(X/W(k))$$

is a finite free $W(k)$-module of rank $2g$, equipped with a σ-semilinear Frobenius endomorphism

$$F : H^1_{\mathrm{cris}}(X/W(k)) \longrightarrow H^1_{\mathrm{cris}}(X/W(k)),$$

where σ denotes the Witt vector Frobenius.

Comparison with de Rham cohomology. Crystalline cohomology reduces modulo p to de Rham cohomology:

$$\boxed{H^1_{\mathrm{cris}}(X/W(k)) \otimes_{W(k)} k \;\cong\; H^1_{\mathrm{dR}}(X).}$$

Under this identification, the Hodge filtration on de Rham cohomology

$$0 \longrightarrow H^0(X, \Omega^1_X) \longrightarrow H^1_{\mathrm{dR}}(X) \longrightarrow H^1(X, \mathcal{O}_X) \longrightarrow 0$$

arises from the natural filtration on crystalline cohomology.

Frobenius and the Cartier operator. Reducing Frobenius modulo p yields an endomorphism of $H^1_{\mathrm{dR}}(X)$, still denoted by F. Its induced action on the Hodge filtration has the following properties:

- F acts trivially on $H^1(X, \mathcal{O}_X)$,
- F induces the Cartier operator

$$C : H^0(X, \Omega^1_X) \longrightarrow H^0(X, \Omega^1_X).$$

In particular,

$$\boxed{\ker(C) = \ker\big(F : H^0(X, \Omega^1_X) \to H^0(X, \Omega^1_X)\big)\,.}$$

5.2.1. *Étale cohomology with $\mathbb{F}_p$-coefficients.* There is a canonical comparison between crystalline and étale cohomology with $\mathbb{F}_p$-coefficients:

$$\boxed{H^1_{ét}(X, \mathbb{F}_p) \;\cong\; \ker\big(F - \mathrm{id} : H^1_{\mathrm{cris}}(X/W(k))/p \to H^1_{\mathrm{cris}}(X/W(k))/p\big)\,.}$$

Via the reduction isomorphism $H^1_{\mathrm{cris}}(X/W(k))/p \cong H^1_{\mathrm{dR}}(X)$, this identifies $H^1_{ét}(X, \mathbb{F}_p)$ with a subspace of $H^1_{\mathrm{dR}}(X)$.

Using the Hodge filtration, one obtains the canonical identification

$$\boxed{H^1_{ét}(X, \mathbb{F}_p) \otimes_{\mathbb{F}_p} k \;\cong\; \ker(C) \;\subset\; H^0(X, \Omega^1_X) \;\subset\; H^1_{\mathrm{dR}}(X).}$$

Ordinary and supersingular curves. The action of Frobenius on crystalline cohomology reflects the geometry of X.

Ordinary case. If X is ordinary, then

$$\dim_{\mathbb{F}_p} H^1_{ét}(X, \mathbb{F}_p) = g,$$

the Cartier operator is bijective on a g-dimensional subspace of $H^0(X, \Omega^1_X)$, and the Hodge filtration splits.

Supersingular case. If X is supersingular, Frobenius acts nilpotently modulo p,

$$H^1_{ét}(X, \mathbb{F}_p) = 0,$$

and the Cartier operator is nilpotent.

Summary

Crystalline cohomology provides a single object encoding both de Rham and étale information:

$$\begin{aligned} H^1_{\mathrm{cris}}(X/W(k)) &\rightsquigarrow H^1_{\mathrm{dR}}(X) \quad (\mathrm{mod}\, p), \\ \ker(F - \mathrm{id}) &\cong H^1_{ét}(X, \mathbb{F}_p), \\ \ker(C) &\cong H^1_{ét}(X, \mathbb{F}_p) \otimes k. \end{aligned}$$

Thus $\mathbb{F}_p$-étale cohomology is precisely the Frobenius-fixed part of crystalline cohomology, and it appears inside de Rham cohomology as the space of Cartier-invariant differentials.

5.2.2. *p-Rank and Representation.* The p-rank of X, denoted h_X, is the dimension of the $\mathbb{F}_p$-vector space $H^1(X_{et}, \mathbb{F}_p)$. This p-rank is defined as the stable rank of the Frobenius morphism F on $H^1(X, \mathcal{O}_X)$.

The representation studied is the dual of the semisimple part of $H^1(X, \mathcal{O}_X)$. Using Serre duality and the identification of the dual map to F as the Cartier operator $\mathcal{C}$ on differentials, the representation is $V_D = H^0(X, \Omega_X(D))^s$, the space of $\mathcal{C}$-semisimple differentials.

(1) V_D is a finite-dimensional representation of G over k.
(2) V_D does not depend on the multiplicities of the G-invariant effective divisor D, only on the reduced divisor D^{red}. Thus, $V_D = V_{D^{red}}$.

5.2.3. *Modular Representation Theory Concepts.* Any representation M of G over k admits a unique decomposition into a non-projective part (core) and a projective part:

$$M \cong \mathrm{core}(M) \oplus \bigoplus_{S \in \mathrm{Irr}\, G} P_G(S)^{\oplus b(M,S)}.$$

(1) $\mathrm{core}(M)$: The sum of all non-projective indecomposable summands.
(2) $P_G(S)$: The projective cover of the simple module S.
(3) $b(M,S)$: The multiplicity of $P_G(S)$ as a direct summand. If $M = V_D$, these multiplicities are called the *Borne invariants* $b(G,D,S)$.

5.2.4. *Determination of the Core.* The core is completely determined by local data related to the fixed point structure of the action. This is surprising since, in general, non-semisimple representations are difficult to classify.

The core module of the p-rank representation V_D is isomorphic to the loop space of a ramification module.
(1) The ramification module $R_{G,D^{red},\tilde{D}}$ is defined using a sufficiently large G-invariant divisor $\tilde{D} \supset D^{red}$.
(2) The core of V_D is isomorphic to the loop space of this ramification module:

$$\mathrm{core}(V_D) \cong C_{D^{red}} := \Omega_G(R_{G,D^{red},\tilde{D}}),$$

where Ω_G is the first loop space operator. The isomorphism class of the core does not depend on the choice of the sufficiently large divisor $\tilde{D}$.

As a corollary, a congruence formula for the p-rank h_X is derived:

$$h_X \equiv 1 - r \pmod{p^n},$$

where r is the number of points of X with non-trivial stabilizer in G, and p^n is the p-part of the order of G.

5.2.5. *Borne Invariants of Quotient Curves.* The second main result relates the Borne invariants of the original curve X to those of a quotient curve $Y = X/N$, where N is a normal subgroup of G. This offers a partial solution for calculating the Borne invariants of X by using invariants of the "smaller" curve Y.

Theorem 5.4 states that the Borne invariants $b(G,T)$ of X with respect to G are related to the Borne invariants $b(H,T)$ of $Y = X/N$ with respect to $H = G/N$ for an irreducible representation $T \in \mathrm{Irr}\, H$ by the formula:

$$b(G,T) + d(G,X,T) = b(H,T) + d(H,Y,T),$$

where $d(G,X,T)$ and $d(H,Y,T)$ are dimensions of certain locally trivial first cohomology groups related to the ramification structure.
(1) This formula generalizes Nakajima's equivariant Deuring-Shafarevich formula.
(2) If N is a p-group, the Borne invariants of Y determine all Borne invariants of X.
(3) If the action is tame (i.e., $p \nmid |G_x|$ for all $x \in X$), the formula simplifies: $d(G,X,S) = \dim H^1(G,S)$.

6. Equivariant Koszul Cohomology of Canonical Curves

The genealogy of the Koszul complex begins with Hilbert's foundational 1890 paper [28], where it appears implicitly in his proof of the famous syzygy theorem that bears his name. However, its formal algebraic structure was not codified until Jean-Louis Koszul's 1950 paper [41]. Though his primary motivation was topological, Koszul's treatment provided the abstract definitions and results that characterize the modern understanding of the complex today.

The modern era of the theory began with Mark Green, who introduced the geometric counterpart of the Koszul complex [20] and demonstrated that geometric properties of projective varieties could be reinterpreted through homological vanishing theorems. Building on this foundation, Green and Lazarsfeld extended several classical results concerning syzygies of projective varieties into the language of Koszul cohomology, see the surveys [21] and [42]. Subsequent breakthroughs eventually resolved influential conjectures on the subject, leading to a period of significant growth for the field; see the preface of [1] for a detailed account of this expansion and its various applications. Despite the depth of these developments, a direction that remains largely unexplored is the extent to which these results extend to the generalized setting where a finite group is permitted to act on the variety.

In [30] the behavior of the Koszul complex and its cohomology in the equivariant setting is investigated, where a finite group G acts on a variety X. The focus lies primarily on the case of smooth projective curves, with particular emphasis on canonical curves.

For a G-equivariant very ample line bundle L on a variety X, the Koszul complex is a complex of kG-modules. The virtual representation of its cohomology, $[K_{p,q}(X, L)]$, is given by:

$$[K_{p,q}(X,L)] = [H^0(\bigwedge^{p} \mathcal{H}_L \otimes L^{\otimes q})] - [H^0(\bigwedge^{p+1} V \otimes L^{\otimes (q-1)})] + [H^0(\bigwedge^{p+1} \mathcal{H}_L \otimes L^{\otimes (q-1)})]$$

where $\mathcal{H}_L$ is the Lazarsfeld bundle (the kernel of the evaluation map) and $V = H^0(L)$.

Suppose that X is a curve of genus $g \geq 4$ and $L = \Omega_X$ is the canonical bundle.To compute the term involving the Lazarsfeld bundle $\mathcal{H}_{\Omega_X}$, the Equivariant Riemann-Roch Theorem can be employed. And an explicit decomposition of the representation into irreducible components based on the ramification data of the cover $\pi : X \to X/G$ is given. The multiplicity of an irreducible representation I in the kernel bundle term is expressed using local multiplicities $n_{k,Q,I}$ and the fractional parts of terms involving the ramification index e_Q. Observe that the term $[\bigwedge^p V \otimes V] - [\bigwedge^{p+1} V]$ corresponds to the Schur functor (or Weyl module) $\mathbb{S}_{(2,1^{p-2})}(V)$. Because the direct decomposition of such modules is notoriously difficult (the plethysm problem), they instead provide generating functions to describe these representations. Theorem 3.5.1 in karagiannis2025equivariantkoszulcohomologycanonical provides a generating function $\mathrm{Sch}_V(t, g)$ for the character values of the Schur module at any group element $g \in G$:

$$\mathrm{Sch}_V(t,g) = \frac{1}{t}\left[(\chi_V(g)t - 1)\prod_{I \in Irr(G)} \prod_{j=1}^{\dim I} (1 + t \cdot \xi_{I,j}(g))^{m_I} + 1\right],$$

where $\xi_{I,j}$ are the eigenvalues of the action of g on the irreducible representation I, and m_I is the multiplicity of I in $V = H^0(\Omega_X)$.

References

[1] Marian Aprodu and Jan Nagel. *Koszul cohomology and algebraic geometry*, volume 52 of *University Lecture Series*. American Mathematical Society, Providence, RI, 2010.

[2] Frauke M. Bleher, Ted Chinburg, and Aristides Kontogeorgis. Galois structure of the holomorphic differentials of curves. *J. Number Theory*, 216:1–68, 2020.

[3] Frauke M. Bleher and Adam Wood. The Galois module structure of holomorphic poly-differentials and Riemann-Roch spaces. *J. Algebra*, 631:756–803, 2023.

[4] Z. I. Borevi ˇc and D. K. Faddeev. Theory of homology in groups. II. Projective resolutions of finite groups. *Vestnik Leningrad. Univ.*, 14(7):72–87, 1959.

[5] Niels Borne. Cohomology of G-sheaves in positive characteristic. *Adv. Math.*, 201(2):454–515, 2006.

[6] Helmut Boseck. Zur Theorie der Weierstrasspunkte. *Math. Nachr.*, 19:29–63, 1958.

[7] Hara Charalambous, Kostas Karagiannis, Sotiris Karanikolopoulos, and Aristides Kontogeorgis. Computational aspects of equivariant hilbert series of canonical rings for algebraic curves, 2022.

[8] Hara Charalambous, Kostas Karagiannis, Sotiris Karanikolopoulos, and Aristides Kontogeorgis. The equivariant hilbert series of the canonical ring of fermat curves. *Indagationes Mathematicae (accepted)*, 2022.

[9] C. Chevalley, A. Weil, and E. Hecke. Über das verhalten der integrale 1. gattung bei automorphismen des funktionenkörpers. *Abh. Math. Sem. Univ. Hamburg*, 10(1):358–361, 1934.

[10] C Chevalley, A Weil, and E Hecke. Über das verhalten der integrale 1. gattung bei automorphismen des funktionenkörpers. *Abh. Math. Semin. Univ. Hambg.*, 1934.

[11] Ted Chinburg and Boas Erez. Equivariant Euler-Poincaré characteristics and tameness. Number 209, pages 13, 179–194. 1992. Journées Arithmétiques, 1991 (Geneva).

[12] Gunther Cornelissen, Fumiharu Kato, and Aristides Kontogeorgis. Discontinuous groups in positive characteristic and automorphisms of Mumford curves. *Math. Ann.*, 320(1):55–85, 2001.

[13] G. Ellingsrud and K. Lønsted. An equivariant Lefschetz formula for finite reductive groups. *Math. Ann.*, 251(3):253–261, 1980.

[14] Hershel M. Farkas and Irwin Kra. *Riemann surfaces*, volume 71 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1980.

[15] Jędrzej Garnek and Aristides Kontogeorgis. The de Rham cohomology of covers with a cyclic p-Sylow subgroup. *J. Algebra*, 687:151–178, 2026.

[16] L Gerritzen. On non-archimedean representations of abelian varieties. *Math. Ann.*, 196(4):323–346, 1 December 1972.

[17] Lothar Gerritzen and Marius van der Put. *Schottky groups and Mumford curves*, volume 817 of *Lecture Notes in Mathematics*. Springer, Berlin, 1980.

[18] M Giulietti and G Korchmáros. Algebraic curves with a large non-tame automorphism group fixing no point. *Trans. Amer. Math. Soc.*, 362(11):5983–5983, 1 November 2010.

[19] Massimo Giulietti and Gábor Korchmáros. Nakajima's remark on Henn's proof. *Electronic Notes in Discrete Mathematics*, 40:135–138, 15 May 2013.

[20] Mark L. Green. Koszul cohomology and the geometry of projective varieties. *J. Differential Geom.*, 19(1):125–171, 1984.

[21] Mark L. Green. Koszul cohomology and geometry. In *Lectures on Riemann surfaces (Trieste, 1987)*, pages 177–200. World Sci. Publ., Teaneck, NJ, 1989.

[22] Alexander Grothendieck. Sur quelques points d'algèbre homologique. *Tôhoku Math. J. (2)*, 9:119–221, 1957.
[23] Robin Hartshorne. *Algebraic Geometry*. Springer-Verlag, New York, 1977. Graduate Texts in Mathematics, No. 52.
[24] A. Heller and I. Reiner. Representations of cyclic groups in rings of integers. I. *Ann. of Math. (2)*, 76:73–92, 1962.
[25] Hans-Wolfgang Henn. Funktionenkörper mit grosser Automorphismengruppe. *J. Reine Angew. Math.*, 302:96–115, 1978.
[26] Frank Herrlich. Die Ordnung der Automorphismengruppe einer p-adischen Schottkykurve. *Math. Ann.*, 246(2):125–130, 1979/80.
[27] D. G. Higman. Indecomposable representations at characteristic p. *Duke Math. J.*, 21:377–381, 1954.
[28] David Hilbert. Ueber die Theorie der algebraischen Formen. *Math. Ann.*, 36(4):473–534, 1890.
[29] Adolf Hurwitz. Über algebraische Gebilde mit eindeutigen Tranformationen in sich. *Math. Ann.*, 41:403–442, 1893.
[30] Kostas Karagiannis, Aristides Kontogeorgis, and Konstantia Manousou Sotiropoulou. Equivariant koszul cohomology of canonical curves, 2025.
[31] Sotiris Karanikolopoulos and Aristides Kontogeorgis. Representation of cyclic groups in positive characteristic and Weierstrass semigroups. *J. Number Theory*, 133(1):158–175, 2013.
[32] Sotiris Karanikolopoulos and Aristides Kontogeorgis. Automorphisms of curves and Weierstrass semigroups for Harbater-Katz-Gabber covers. *Trans. Amer. Math. Soc.*, 371(9):6377–6402, 2019.
[33] A. Karrass, A. Pietrowski, and D. Solitar. Finite and infinite cyclic extensions of free groups. *J. Austral. Math. Soc.*, 16:458–466, 1973. Collection of articles dedicated to the memory of Hanna Neumann, IV.
[34] George Katsimprakis and Aristides Kontogeorgis. On cyclic groups covers of the projective line, 2025.
[35] Bernhard Köck. Galois structure of Zariski cohomology for weakly ramified covers of curves. *Amer. J. Math.*, 126(5):1085–1107, 2004.
[36] Bernhard Köck. Computing the equivariant Euler characteristic of Zariski and étale sheaves on curves. *Homology Homotopy Appl.*, 7(3):83–98, 2005.
[37] Bernhard Köck and Aristides Kontogeorgis. Quadratic differentials and equivariant deformation theory of curves. *Ann. Inst. Fourier (Grenoble)*, 62(3):1015–1043, 2012.
[38] Aristides Kontogeorgis. Automorphisms of Fermat-like varieties. *Manuscripta Math.*, 107(2):187–205, February 2002.
[39] Aristides Kontogeorgis and Victor Rotger. On abelian automorphism groups of Mumford curves. *Bull. Lond. Math. Soc.*, 40(3):353–362, 2008.
[40] Aristides Kontogeorgis and Dimitra-Dionysia Stergiopoulou. On the galois-module structure of polydifferentials of artin-schreier-mumford curves, modular and integral representation theory, 2025.
[41] J. L. Koszul. Sur un type d'algèbres différentielles en rapport avec la transgression. In *Colloque de topologie (espaces fibrés), Bruxelles, 1950*, pages 73–81. Georges Thone, Liège, 1951.
[42] Robert Lazarsfeld. A sampling of vector bundle techniques in the study of linear series. In *Lectures on Riemann surfaces (Trieste, 1987)*, pages 500–559. World Sci. Publ., Teaneck, NJ, 1989.
[43] J. Lehner. *Discontinuous Groups and Automorphic Functions*. Mathematical Surveys and Monographs. American Mathematical Society, 1964.
[44] Claus Lehr and Michel Matignon. Automorphism groups for p-cyclic covers of the affine line. *Compos. Math.*, 141(5):1213–1237, 2005.
[45] Heinrich-Wolfgang Leopoldt. Über die Automorphismengruppe des Fermatkörpers. *J. Number Theory*, 56(2):256–282, 1996.
[46] Pedro Ricardo López-Bautista and Gabriel Daniel Villa-Salvador. On the Galois module structure of semisimple holomorphic differentials. *Israel J. Math.*, 116:345–365, 2000.
[47] Daniel J. Madden and Robert C. Valentini. The group of automorphisms of algebraic function fields. *J. Reine Angew. Math.*, 343:162–168, 1983.
[48] Yu. Manin and V. Drinfeld. Periods of p-adic Schottky groups. *J. Reine Angew. Math.*, 262/263:239–247, 1973. Collection of articles dedicated to Helmut Hasse on his seventy-fifth birthday.
[49] Sophie Marques and Kenneth Ward. Holomorphic differentials of certain solvable covers of the projective line over a perfect field. *Math. Nachr.*, 291(13):2057–2083, 2018.
[50] M. Matignon and M. Rocher. On smooth curves endowed with a large automorphism$p$-group in characteristic$p > 0$. *Algebra Number Theory*, 2:887–926, 2008.
[51] P. Monsky. The automorphism groups of algebraic curves. *Ph.D. Thesis University of Chicago*, 1962.
[52] David Mumford. An analytic construction of degenerating curves over complete local rings. *Compositio Math.*, 24:129–174, 1972.
[53] David Mumford. *Abelian varieties*, volume 5 of *Tata Institute of Fundamental Research Studies in Mathematics*. Published for the Tata Institute of Fundamental Research, Bombay; by Hindustan Book Agency, New Delhi, 2008. With appendices by C. P. Ramanujam and Yuri Manin, Corrected reprint of the second (1974) edition.
[54] S. Nakajima. On Galois module structure of the cohomology groups of an algebraic variety. *Invent. Math.*, 75(1):1–8, 1984.
[55] Shōichi Nakajima. Galois module structure of cohomology groups for tamely ramified coverings of algebraic varieties. *J. Number Theory*, 22(1):115–123, 1986.
[56] Shōichi Nakajima. On abelian automorphism groups of algebraic curves. *J. London Math. Soc. (2)*, 36(1):23–32, 1987.
[57] Shōichi Nakajima. p-ranks and automorphism groups of algebraic curves. *Trans. Amer. Math. Soc.*, 303(2):595–607, 1987.
[58] Bjorn Poonen. Varieties without extra automorphisms I: Curves. *Math. Res. Lett.*, 7:67–76, 2000.
[59] Herbert Popp. The singularities of the moduli schemes of curves. *J. Number Theory*, 1(1):90–107, 1 January 1969.
[60] Mike Prest. Wild representation type and undecidability. *Comm. Algebra*, 19(3):919–929, 1991.
[61] Magali Rocher. Large p-groups actions with $|g|/g^2 > 4/(p^2-1)^2$. 22 April 2008.
[62] Magali Rocher. Large p-group actions with a p-elementary abelian derived group. *J. Algebra*, 321(2):704–740, 2009.
[63] Martha Rzedowski-Calderón, Gabriel Villa-Salvador, and Manohar L. Madan. Galois module structure of holomorphic differentials in characteristic p. *Arch. Math. (Basel)*, 66(2):150–156, 1996.

[64] Hermann Ludwig Schmid. Über die automorphismen eines algebraischen funktionenkörpers von primzahlcharakteristik. *J. Reine Angew. Math.*, 179:5–15, 1938.
[65] P. Schneider. Rigid-analytic L-transforms. In *Number theory, Noordwijkerhout 1983 (Noordwijkerhout, 1983)*, volume 1068 of *Lecture Notes in Math.*, pages 216–230. Springer, Berlin, 1984.
[66] Nicolas Stalder. On p-rank representations. *J. Algebra*, 280(2):825–841, 2004.
[67] Henning Stichtenoth. Über die Automorphismengruppe eines algebraischen Funktionenkörpers von Primzahlcharakteristik. I. Eine Abschätzung der Ordnung der Automorphismengruppe. *Arch. Math. (Basel)*, 24:527–544, 1973.
[68] Henning Stichtenoth. Über die Automorphismengruppe eines algebraischen Funktionenkörpers von Primzahlcharakteristik. II. Ein spezieller Typ von Funktionenkörpern. *Arch. Math. (Basel)*, 24:615–631, 1973.
[69] Henning Stichtenoth. *Algebraic function fields and codes*, volume 254 of *Graduate Texts in Mathematics*. Springer-Verlag, Berlin, second edition, 2009.
[70] Tuneo Tamagawa. On unramified extensions of algebraic function fields. *Proc. Japan Acad.*, 27:548–551, 1951.
[71] Jeremy T. Teitelbaum. Values of p-adic L-functions and a p-adic Poisson kernel. *Invent. Math.*, 101(2):395–410, 1990.
[72] Robert C. Valentini. Representations of automorphisms on differentials of function fields of characteristic p. *J. Reine Angew. Math.*, 335:164–179, 1982.
[73] Robert C. Valentini and Manohar L. Madan. Automorphisms and holomorphic differentials in characteristic p. *J. Number Theory*, 13(1):106–115, 1981.